

Acute Effects of Transcranial Photobiomodulation On Glymphatic CSF Dynamics in Two Patients with Amnestic Mild Cognitive Impairment or Early Alzheimer Disease

Carlos Lohmann¹, Puneet Velidi², Guillermo Gonzalez-Garibay³, Suk-Tak Chan⁴, and Paolo Cassano⁵

¹ Department of Psychiatry, Cambridge Health Alliance, Cambridge, MA, USA

² Department of Mathematics and Statistics, University of Victoria, Victoria, BC, Canada

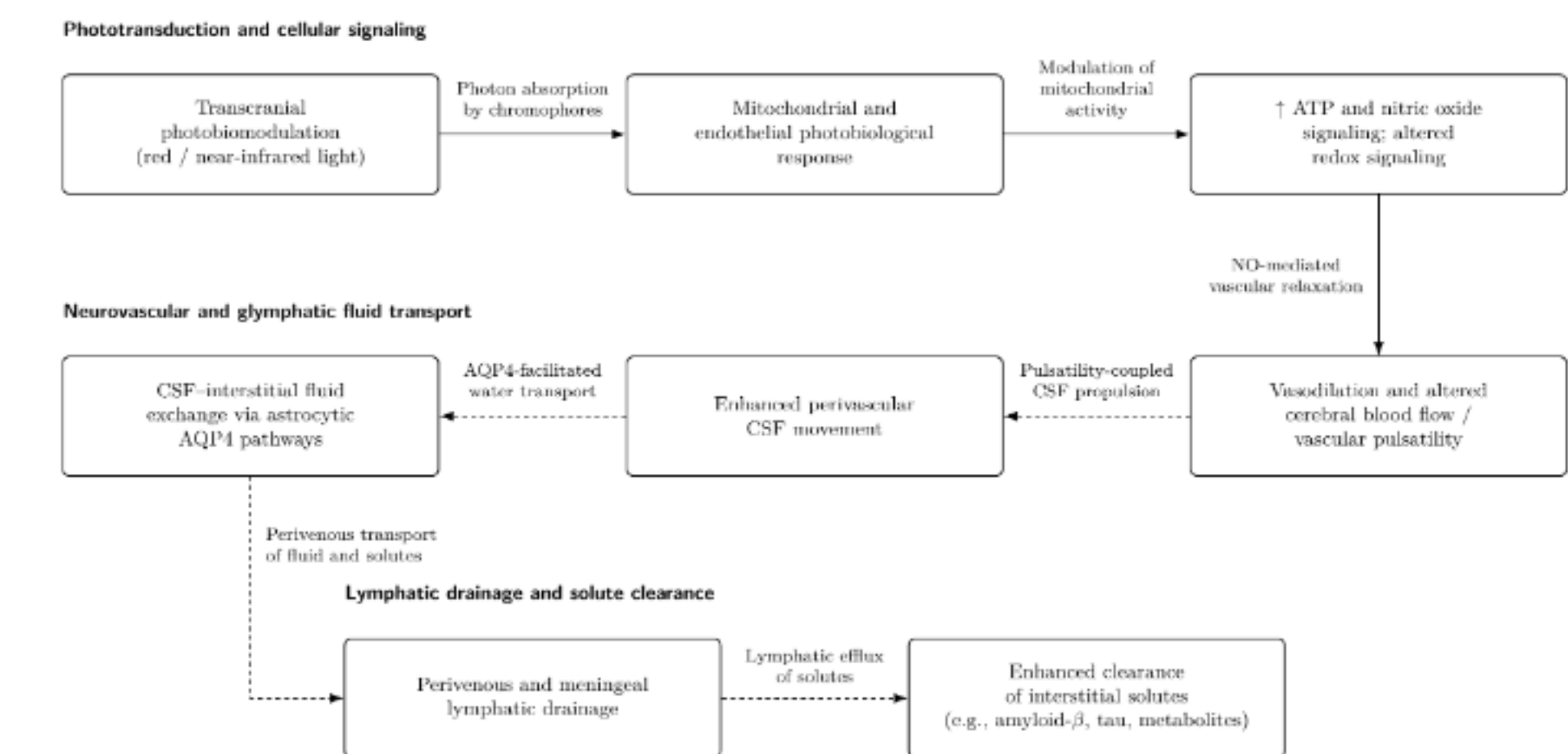
³ Medical School, Universidad Anáhuac México, Mexico City, Mexico

⁴ Athinoula A. Martinos Center for Biomedical Imaging, Department of Radiology, Massachusetts General Hospital, Charlestown, MA, USA

⁵ Department of Psychiatry, Massachusetts General Hospital, Boston, MA, USA

INTRODUCTION

The glymphatic system has attracted interest as a potential therapeutic target because of its role in brain solute clearance and the association between altered BOLD–CSF coupling and Alzheimer disease–related pathology [1,2]. This is relevant to amnestic mild cognitive impairment and early Alzheimer disease, in which pathological changes may precede clinical decline [3]. Transcranial photobiomodulation (tPBM) can influence mitochondrial and cerebrovascular physiology [4], while hemodynamic fluctuations are coupled to CSF dynamics [5]. Recent pulsed-PBM findings and mechanistic proposals motivate investigation in patient populations [6,7]. We examined fourth-ventricular fMRI fluctuations and GM BOLD–CSF coupling using complementary multi-echo and fast acquisitions in two TRAP-AD participants' first study visit.



METHODS

Subject 1 received bilateral prefrontal 808-nm continuous-wave tPBM; Subject 2 received sham. This poster is restricted to each participant's first visit: six multi-echo acquisitions, comprising protocol-labeled pre/high/post runs in each subject. The protocols were analyzed separately; visit labels were not treated as stimulation phases. Multi-echo images (TR 2.522 s; TE 12.80/32.34/51.86 ms; 240 volumes per run) underwent slice-timing correction and shared first-echo-derived motion correction. Tedana 26.0.3 provided optimally combined, ME-ICA-denoised GM BOLD. Anatomical GM masks were registered to each visit's SBRef; fourth-ventricle masks were manually defined. CSF signals were extracted separately from the shortest echo, without ME-ICA; other echoes and non-ICA optimal combination were sensitivity analyses. Four first-visit fast single-echo multiband EPI acquisitions were obtained on a 3-T Siemens MAGNETOM Prisma Fit scanner (TR/TE 378/31 ms; flip angle 37°; 2.5-mm isotropic voxels; 40 slices; multiband factor 8; 951–952 volumes, approximately 6 min/run). Fast data underwent motion correction, without slice-timing correction or ME-ICA; GM and fourth-ventricular signals were extracted separately from each acquisition. Both protocols used quadratic detrending, 24-term motion regression, initial stabilization exclusion and padded censoring at FD >0.5 mm [9], without global-signal regression. CSF signal amplitude was the temporal SD of the unfiltered, nuisance-regressed fourth-ventricular ROI-mean signal. Welch spectra used uninterrupted Hann windows with 50% overlap: 64 volumes for multi-echo and 384 for fast fMRI [8]. Slow-band power covered 0.03–0.07 Hz. Same-acquisition BOLD–CSF correlations used joint 0.01–0.10 Hz filtering/nuisance removal within the longest eligible continuous segment [2,5]. Fixed CSF delays were 2.522 s for multi-echo and 3.024 s for fast fMRI; approximately 120-s rolling correlations were descriptive. Coupling required at least 240 s of continuous retained data. Subject 2 pre/post time courses were displayed separately as unfiltered censored residuals, without coupling estimates.

DISCUSSION AND CONCLUSION

The CW-tPBM participant showed concordant pre-to-post increases in fourth-ventricular CSF signal variability across multi-echo (+9.3%) and fast (+10.1%) acquisitions, alongside a 49.0% increase in multi-echo slow-band power. Together, these amplitude and spectral changes provide an encouraging preliminary imaging signal of altered CSF-related oscillatory dynamics. These findings motivate further investigation of whether tPBM influences CSF movement and its relationship with cerebral hemodynamics, using larger controlled studies and direct flow-sensitive measurements.

ACKNOWLEDGEMENTS

This research is funded by National Institutes of Health–National Institute on Aging (NIH–NIA), grant number: R01AG068248, and the Alzheimer's Association, grant number: PTCG-20-706555.

REFS



RESULTS

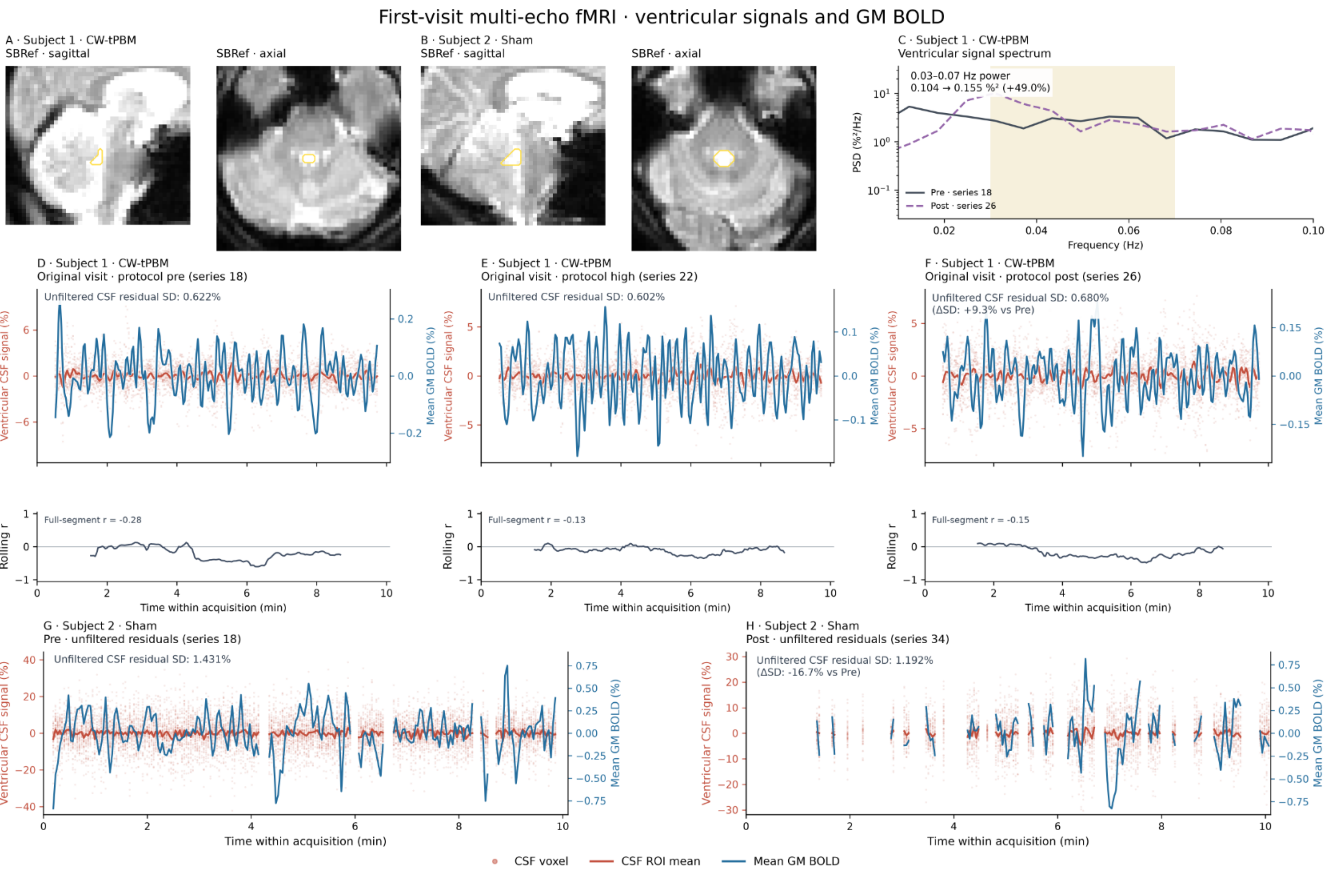


FIGURE 1. FIRST-VISIT MULTI-ECHO VENTRICULAR SIGNALS AND GM BOLD A–B: native-SBRef sagittal/axial fourth-ventricle masks (yellow) for Subject 1 (CW-tPBM) and Subject 2 (sham). C: Subject 1 shortest-echo pre/post ventricular spectra; shading marks 0.03–0.07 Hz. D–F: Subject 1 protocol-pre/high/post band-limited CSF voxel scatter and ROI mean (red, left axis), GM BOLD mean only (blue, right axis), and rolling correlations. GM uses tedana-denoised three-echo data; CSF-region signal uses echo 1 without ME-ICA. Subject 1 coupling uses joint 0.01–0.10-Hz filtering/nuisance removal, a fixed 2.522-s ventricular-signal delay and 49-sample (~124-s) rolling windows [2,5]. G–H restore Subject 2's Pre and Post time courses as UNFILTERED nuisance residuals, not band-limited coupling curves. Motion censoring is unchanged: FD >0.5 mm, one neighboring volume on each side, and the initial four volumes are excluded. Gaps are retained, without interpolation or connecting lines across censored frames. Pre/Post retain 214/240 and 103/240 volumes. Unfiltered CSF residual SD is 1.431% versus 1.192% (–16.7%, calculated from unrounded values); unequal censoring and residual artifacts limit interpretation. No Subject 2 coupling curve, rolling correlation or spectrum is restored: the longest retained segments remain 239.6 s Pre, 45.4 s High and 42.9 s Post, below the unchanged 240-s coupling requirement; pre/post spectra also fail the existing spectral QC. The High run is not added as a descriptive endpoint panel. Missing estimates are not zero [9]. All SD annotations describe the temporal SD of the UNFILTERED, nuisance-regressed CSF ROI mean, not GM SD or the SD of a band-limited plotted curve. Subject 1's pre/post CSF SD is 0.622%/0.680% (+9.3%). Spectra use uninterrupted 64-volume Hann windows with 50% overlap [8]. The internal fourth-ventricle ROI is not an inflow-optimized acquisition boundary; inflow specificity is unvalidated, and ventricular signal changes do not directly measure flow or glymphatic clearance [5].

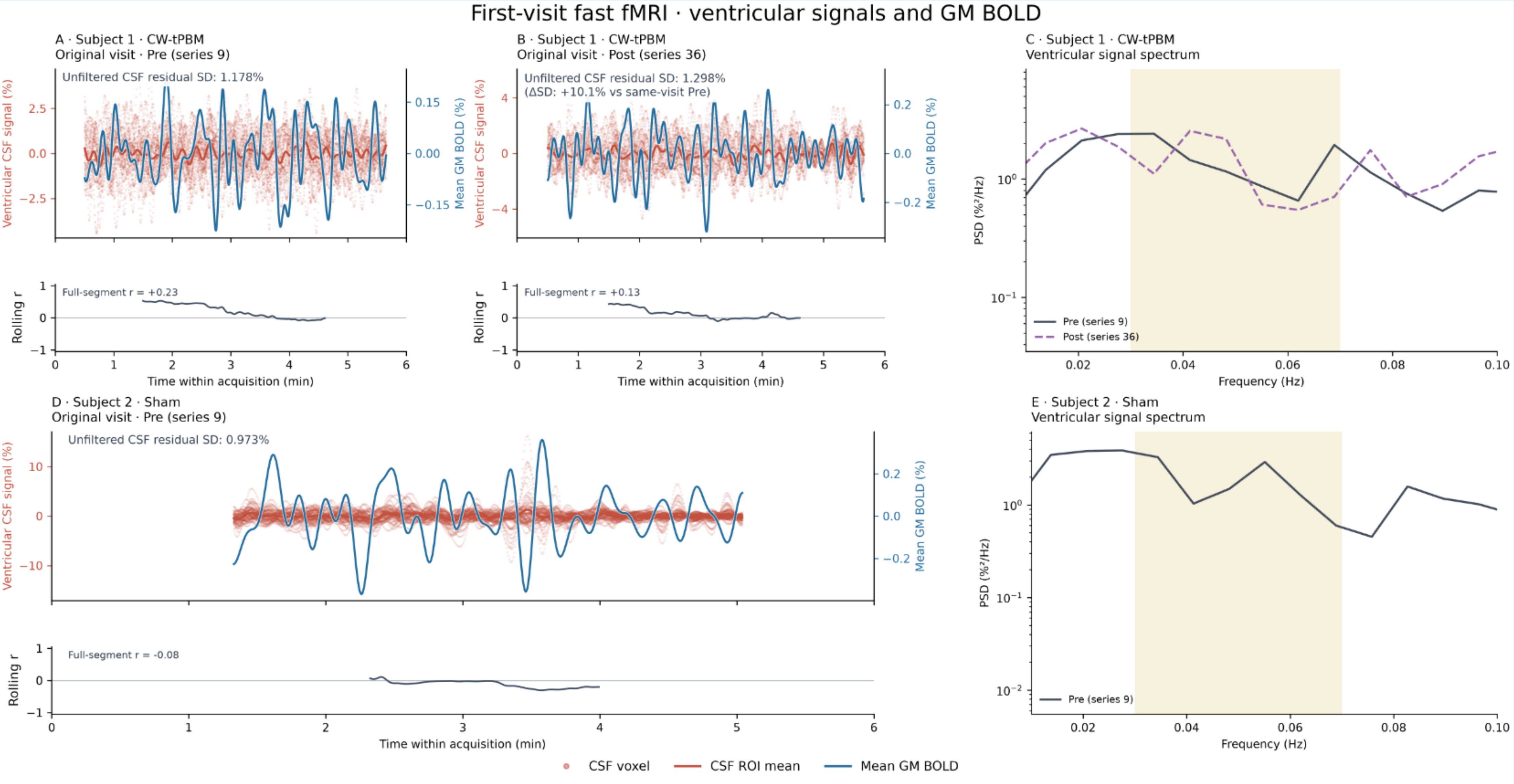


FIGURE 2. FIRST-VISIT FAST VENTRICULAR SIGNALS AND EXPLORATORY GM BOLD COUPLING A–B: Subject 1 Pre/Post; D: Subject 2 Pre. CSF voxel scatter and ROI mean are red (left axis); GM BOLD mean only is blue (right axis). C and E: corresponding ventricular spectra, with 0.03–0.07 Hz shaded. Later visits are excluded from this poster. Subject 1's unfiltered CSF residual SD changes from 1.178% to 1.298% (+10.1%); Subject 2 Pre is 0.973%. Subject 2 original-visit fast series 48 remains unavailable because the fourth-ventricle ROI lies outside its acquired coverage, so a fast pre/post SD change cannot be estimated. Fast processing and results are unchanged: quadratic detrending, 24-term motion regression, initial 27-volume exclusion and FD >0.5-mm censoring with three preceding/six following volumes excluded. Coupling uses joint 0.01–0.10-Hz filtering/nuisance removal, the longest retained continuous segment (at least 240 s), ~20-s edge trimming, a fixed 3.024-s ventricular-signal delay and 317-sample (~120-s) rolling windows. Full-segment correlations use the same lag. Spectra use unfiltered residuals and uninterrupted 384-volume Hann windows with 50% overlap [2,8,9]. Without fast slice-timing correction, lag values are nominal.

Subject 1: shortest-echo CSF SD was 0.622% in protocol-pre and 0.680% in protocol-post (+9.3%). Power at 0.03–0.07 Hz was 0.104 versus 0.155 %² (+49.0%); its fraction of 0.01–0.10 Hz power was 46.1% versus 53.9%. At a +2.522-s CSF delay, ME BOLD–CSF r was –0.275 versus –0.148. CSF amplitude changes were echo-dependent (–1.3% to +22.1% across echoes). First-visit fast CSF SD increased from 1.178% to 1.298% (+10.1%). Subject 2: shortest-echo CSF SD was 1.431% in protocol-pre and 1.192% in protocol-post (–16.7%). A pre/post slow-band power contrast was not estimable under the continuous-data QC requirements. A pre/post coupling contrast was not estimable under the continuous-data QC requirements. After exclusions, 214/240 pre and 103/240 post volumes remained; unequal censoring limits interpretation of the SD difference. Multi-echo coupling was estimable in 3/6 first-visit acquisitions, all from Subject 1. Fast CSF signals were available in 3/4 first-visit acquisitions; Subject 2's original-visit series 48 lacked CSF ROI coverage. Motion-regressed fast correlations at +3.024 s ranged from –0.083 to +0.233, without a consistent fixed-lag anticorrelation.